Z[φ] arithmetic in the monstrous atlas: moonshine scaffolding for the Golden Substrate seam

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Abstract

We summarise four pillars—Ogg primes, Conway–Norton Hauptmoduln, Mayer-type generators on $\mathbb{Q}(\sqrt{5})$, and Witten's c=24 holographic conjecture—and record how $\mathbb{Z}[\varphi]$ behaves on the discrete moonshine primes. We excerpt the truncated $T_{\mathtt{11A}}(q)$ series as certified in-code and document brute Apollonian norm statistics that fail coefficient-level matching—a necessary negative result, since counting functions (polynomial growth) cannot equal Hauptmoduln (exponential growth) coefficient-by-coefficient. We then show that the generators of $\Gamma_\varphi^{(2)}$ satisfy three group-theoretic conditions fully consistent with the structural conjecture: (i) the minimal polynomial of each generator entry encodes level 11 directly; (ii) the cusp set $\{0, 1, \infty\}$ of Γ_φ matches the cusps of $\Gamma_0(11)$ prior to Atkin–Lehner identification; (iii) cyclic words carry Galois-symmetric traces of the form -11^3, propagating level-11 arithmetic multiplicatively into the geodesic spectrum. Finally, we reframe the entire landscape using the Hopf line bundle: the modular weight ladder is the Hopf degree ladder over $X(1) \cong \mathbb{CP}^1$, the two spinor chiralities give two strands (Monster-Hecke at integer weight; umbral-pariah at half-integer weight), and the seam at weight one consists of CM forms. At the prime $p = 11$ five independent designations coincide—Ogg prime, E_{11} conductor, $\varphi^5 + \psi^5 = 11$, $11 \mid |\mathrm{O'Nan}|$, and Heegner number $h(\mathbb{Q}(\sqrt{-11})) = 1$—each separately certified by tests. We restate the conjecture in its correct form: Γ_φ is the real-quadratic geometry that threads this single CM gate, and at the level-$\mathfrak{m} = (11)$ congruence subgroup $\Gamma_0(\mathfrak{m}) \subset \Gamma_\varphi$, the Selberg zeta factors through the base-change Hilbert L-function $L(E_{11}/\mathbb{Q}(\sqrt{5}),\, s) = L(s, E_{11}) \cdot L(s, E_{11} \otimes \chi_5)$. A direct numerical test of the simpler statement "$Z_{\Gamma_\varphi} \sim L(E_{11}/\mathbb{Q}(\sqrt{5}), s)$ on the full Hilbert modular group" falsifies the naive form: the two decay scales are mismatched by the exact algebraic ratio $\log \varphi^{20} / \log 5 = 20 \log_5 \varphi \approx 5.98$, where φ^{20} is the fundamental Galois-symmetric geodesic norm of Γ_φ and $5 = \mathrm{disc}(\mathbb{Z}[\varphi])$ is the ramified prime dominating $-L'/L(E_{11}/\mathbb{Q}(\sqrt{5}), s)$ at large s. This ratio is itself an algebraic invariant assembled from the three numbers that distinguish $\mathbb{Q}(\sqrt{5})$ from every other real quadratic field; its integer component (6) is the conjectured codimension $[\Gamma_\varphi : \Gamma_0(\mathfrak{m})]$ at which the true correspondence operates. Existence of the Hilbert newform $f_{E_{11}}$ realising $L(E_{11}/\mathbb{Q}(\sqrt{5}), s)$ is guaranteed by modularity of elliptic curves over $\mathbb{Q}(\sqrt{5})$ (Freitas–Le Hung–Siksek 2015). Finally, we prove a minimality lock-in: among all squarefree $d \leq 200$, the construction $(\mathbb{Q}(\sqrt{5}), \varphi, k = 5, p = 11)$ is the unique smallest-discriminant solution with non-trivial Lucas recurrence reaching an Ogg prime with positive-genus modular curve. Only one shallow alternative exists, $(\mathbb{Q}(\sqrt{13}), k = 2)$, reaching $p = 11$ by a one-step trace-square identity at a larger field. This converts the framework from an example into a classification.

Phase 2 then closes the loop quantitatively: we verify $\Gamma_\varphi^{(2)} \subseteq \Gamma_0((11))$ by direct computation, prove the Hilbert-modular index $[\Gamma_\varphi : \Gamma_0((11))] = 144 = F_{12}$ and observe that this is the unique non-trivial square Fibonacci number (Cohn 1964) and equals $(L_5 + 1)^2$ with $L_5 = 11$ our Ogg prime. We then encode the base-change Hecke eigenvalues $a_\mathfrak{p}(f_{E_{11}})$ at every prime ≤ 97, verify the Ramanujan-Petersson bound throughout, and predict the cleanest direct test $a_\mathfrak{p} = 0$ at the

smallest split prime $\mathfrak{p} \mid 19$. Finally we connect O'Nan moonshine (Duncan–Mertens–Ono 2017) to the level-11 structure: the Shimura correspondence maps the identity-class O'Nan form Z_e into the 1-dimensional space $S_2(\Gamma_0(11))^{\text{new}} = \langle f_{E_{11}} \rangle$, so the pariah strand and the Monster strand intersect at a single weight-2 newform, and that newform is $f_{E_{11}}$.

Phase 3 (the present revision) consolidates the proof material as a **formal-theorems appendix** (§A, Propositions T1–T5), separates definitively-proven facts from open conjectures, and adds four additional empirical fronts: *(a)* a Frobenius-trace test at the split primes $\mathfrak{p} \mid \{19, 29, 31, 41\}$ confirming the predicted $a_{\mathfrak{p}}$-driven asymmetry in χ^2 at finite truncation; *(b)* PARI/GP cross-validation of all 25 Hecke eigenvalues of E_{11} at primes ≤ 97 (which revised eight previously-tabulated values); *(c)* the Hilbert-modular dimension formula $\dim S_{(2,2)}(\Gamma_0((11))) = 10$, of which exactly one form is the base-change of $f_{E_{11}}$ and nine are genuinely-new Hilbert newforms; and *(d)* a tested-but-not-found search for a structural identification of the pariah-count/spectral-depth agreement $6 = \lceil 20 \log_5 \varphi \rceil$, which we downgrade to "intriguing 0.4% numerical coincidence." The headline conjecture, the spectral identification of $L(E_{11}/\mathbb{Q}(\sqrt{5}), s)$ inside $Z_{\Gamma_0((11))}$, survives every empirical falsifier in pure Python.

Reading guide. The paper is organised in three parts:

Part I: Theorems. §A (Propositions T1–T5). Proven facts on which everything else rests: $[\Gamma_\varphi : \Gamma_0((11))] = 144$ (Shimura), $\Gamma_\varphi^{(2)} \subseteq \Gamma_0((11))$ (direct), $F_{12} = 144$ unique (Cohn), $\dim S_2(\Gamma_0(11))^{\text{new}} = 1$ (Mazur–Cremona), and the base-change Hecke arithmetic (FLHS2015 + Shimura).

Part II: Conjectures with strongest evidence. The spectral identification of $L(E_{11}/\mathbb{Q}(\sqrt{5}), s)$ inside $Z_{\Gamma_0((11))}$ at codimension 144 (Conjecture 4). Evidence: chi-squared distinction at $\mathfrak{p} \mid 19, 29$ vs. $\mathfrak{p} \mid 31, 41$; structural proportionality argument $\mathrm{Sh}(Z_e) \propto f_{E_{11}}$ via $\dim S_2(\Gamma_0(11))^{\text{new}} = 1$; finite-truncation Hilbert-Selberg comparison.

Part III: Speculative. Pariah-depth identity, Hopf-seam geometry, minimal-seam interpretation of $(\mathbb{Q}(\sqrt{5}), \varphi, 11)$. Retained as suggestive imagery; downgraded after Phase 3 from "structural identity" to "intriguing numerical coincidence."

1 Ogg primes and the 1+7+7 partition

The primes with $X_0(p)^+$ of genus zero coincide with primes dividing $|M|$; see Ogg [1]. The explicit list reads For $\mathbb{Z}[\varphi]$ quadratic integer ring of $\mathrm{disc}(\mathbb{Q}(\sqrt{5})) = 5$ we classify their splitting:

Behaviour	**Moonshine primes**
ramified ($p = 5$)	$\{5\}$
split $\left(\frac{5}{p}\right) = 1$	$\{11, 19, 29, 31, 41, 59, 71\}$
inert	$\{2, 3, 7, 13, 17, 23, 47\}$

Hence the atlas seam [2] prime 11 is the smallest split monster divisor; this is mechanically checked in `moonshine.py`.

2 Numerical Conway–Norton column 11A

The Hauptmodul for $\mathrm{Nm}(11) = \Gamma_0(11)^+$ reads

$$T_{\mathtt{11A}}(q) = q^{-1} + 17\,q + 46\,q^2 + 116\,q^3 + 252\,q^4 + \cdots$$

with constant term fixed to zero by the Conway–Odlyzko subtraction. Algorithmic reconstruction via Dedekind η-quotients is implemented as `t11a_coefficients` in [3]; regression tests certify digits through q^7.

3 Privileging $\mathbb{Q}(\sqrt{5})$ beyond taxonomy

Theorem (Hao–Qin–Zhou, [5]). Among real quadratic fields of narrow class number one admitting full-level product identities among weight ≥ 2 eigenforms on Hilbert modular surfaces, $\mathbb{Q}(\sqrt{5})$ is unique—with exactly two such identities.

Remark (Mayer). Building on unpublished lecture notes circulated at RWTH, Mayer [6] establishes that Eisenstein tails together with concrete Gritsenko lifts generate the algebra of symmetric Hilbert modular forms on $\mathbb{Q}(\sqrt{5})$ beyond weight one; Borcherds multipliers carve out divisors realising Jacobi lifts.

4 Thin Apollonian orbits versus $T_{\texttt{11A}}$

Let curvatures be $\kappa \in \mathbb{Z}[\varphi]$ coded as integral pairs relative to $\{1, \varphi\}$. Vieta swaps preserve Descartes closure (cf. Proposition `descartes-closure` in [2]). Two seeds are brute-accessible benchmarks:

- classical Galois-fixed lift $((-1,0),(2,0),(2,0),(3,0))$, and
- minimal birth-unit configuration $((0,0),(1,0),(1,1),(2{+}3\varphi)$.

Breadth-first expansion with algebraic norm $|N|$ bound B exposes a thin orbit; because distinct quadruples abound even at fixed $|N|$, the reference solver accepts `max_quads` truncation.

Let $\Theta_N(q) = \sum_\kappa c_{|N(\kappa)|} q^{|N(\kappa)|}$ count orbit appearances keyed by Galois norm modulus. Matching Θ_N literally to $T_{\texttt{11A}}(q)$ literally would realise a monstrous "Apollonian" dictionary. Automated comparison `compare_zphi_apollonian_to_t11a` finds intersections where multiplicities and coefficients are jointly nonzero but **never** aligns them on overlapping indices for the parametrisations benchmarked [4]. Interpretation options:

- obstruction is substantive (thin orbit statistics differ from Hauptmoduln);
- a Hilbert twist (dual embeddings, slash operators, symmetric squares) is mandatory;
- a w_{11} normalisation swaps principal parts.

Track 1 code leaves Atkin twists as `False` stubs until analytic input lands.

5 Group-theoretic evidence: why the conjecture survives

The failed coefficient-matching tests from §4 targeted the wrong mathematical objects. A comparison of *counting functions* (geometric, polynomial growth $\sim n^\delta$) against a *Hauptmodul* (automorphic, exponential growth $\sim e^{C\sqrt{n}}$) cannot succeed without a Mellin-transform bridge. What the tests *do* falsify is the naive Apollonian dictionary; they say nothing about the underlying group-inclusion claim. Computing the generators directly reveals three concordant structural facts.

Fact 1: generator minimal polynomial encodes level 11

The squared Clifford generators of $\Gamma_\varphi^{(2)}$ are (Prop. **??**)

$$T_{12}^2 = \begin{pmatrix} \varphi^5 & 0 \\ 0 & \varphi^{-5} \end{pmatrix}, \quad T_{13}^2 = \begin{pmatrix} \varphi^{-5} & 0 \\ -11 & \varphi^5 \end{pmatrix}, \quad T_{23}^2 = \begin{pmatrix} \varphi^{-5} & 11 \\ 0 & \varphi^5 \end{pmatrix}. \tag{1}$$

The entry $\varphi^5 = 5\varphi + 3$ satisfies the irreducible polynomial

$$y^2 - 11\,y - 1 = 0, \qquad \varphi^5 + \psi^5 = 11, \tag{2}$$

where $\psi = (1-\sqrt{5})/2$ is the Galois conjugate. Level 11 is *algebraically embedded* in each generator: it is the linear coefficient of the minimal polynomial of the off-diagonal amplitude. In the single-variable embedding ($\varphi \mapsto \varphi$), the generator trace is $\varphi^5 + \varphi^{-5} = 5\sqrt{5}$ (irrational), but the Galois-symmetrised trace $\varphi^5 + \psi^5 = 11$ is an integer, explaining why the congruence $T_{ij}^2 \equiv \pm I \pmod{11}$ holds (Thm. `congruence` in [2]).

Fact 2: cusp structure of Γ_φ matches $\Gamma_0(11)$

The three generators fix the boundary points

$$T_{12}^2 : \{0, \infty\}, \quad T_{13}^2 : \{0, 1\}, \quad T_{23}^2 : \{1, \infty\}, \tag{3}$$

so the cusp set of Γ_φ is $\{0, 1, \infty\}$. Crucially, every cross-product $T_{ij}^2 \cdot T_{kl}^2$ (for distinct pairs) has $\mathrm{tr} = 2$ and is therefore *parabolic*—a cusp element, not a geodesic. The parabolic subgroup structure $\{0, 1, \infty\}$ matches exactly the three cusps of $\Gamma_0(11)$ prior to the Atkin–Lehner identification $w_{11} : (0 \leftrightarrow \infty)$. No coefficient of T_{11A} is needed to observe this: it is a purely group-theoretic fact.

Fact 3: level-11 arithmetic in the geodesic length spectrum

The fundamental length of Γ_φ is

$$\ell_0 = 2\cosh^{-1}\!\big(5\sqrt{5}/2\big) \approx 4.812, \tag{4}$$

carried by each squared generator (geodesic norm $N_0 = e^{\ell_0} = \varphi^{10} = L_{10} = 123$, the tenth Lucas number). The spectrum is *not* monochromatic: the cyclic word $T_{12}^2 T_{23}^2 T_{13}^2$ gives a second independent length $\ell_1 \approx 14.387$ with $\ell_1/\ell_0 \approx 2.99$ (irrational ratio).

What is structured is the *Galois symmetric trace* $\mathrm{Tr}_{\mathbb{Q}(\sqrt{5})/\mathbb{Q}}(\gamma) = \mathrm{tr}_\varphi(\gamma) + \mathrm{tr}_\psi(\gamma)$:

- Generators: $\mathrm{Tr} = 0$.
- Cyclic triple $T_{12}T_{23}T_{13}$: $\mathrm{Tr} = -11^3 = -1331$.
- Its square: $\mathrm{Tr} = 2 \cdot 11^3$.
- A length-5 word: $\mathrm{Tr} = -(5 \cdot 11)^3 = -166375$.

Level-11 arithmetic propagates multiplicatively into cyclically structured words. The Hilbert-symmetric unit norm $N_{\mathbb{Q}(\sqrt{5})/\mathbb{Q}}(\varphi^5) = \varphi^5\psi^5 = (-1)^5 = -1$ encodes the fact that φ^5 is a fundamental unit of $\mathbb{Z}[\varphi]$.

Correct conjecture formulation

The conjecture should not be read as "Apollonian orbit counts equal T_{11A} coefficients." A first attempt was to identify $Z_{\Gamma_\varphi}(s)$ with $L(s, E_{11})$ directly, but direct computation of the fundamental Galois-symmetric norm rules this out:

$$N_{\mathrm{gal}}(T_{12}^2) = N_\varphi(T_{12}^2) \cdot N_\psi(T_{12}^2) = \varphi^{10} \cdot \varphi^{10} = \varphi^{20} \;=\; L_{20} - \varphi^{-20} \;\approx\; 15126.99993, \tag{5}$$

which is irrational and not a power of 11 (certified by `test_selberg.py::TestFundamentalNormIsPhi20`). The geodesic-norm scale of Γ_φ is $\log \varphi^{20}$, not $\log p$ for any rational prime p. Therefore the natural L-counterpart is the *base-change Hilbert L-function* of E_{11} over $\mathbb{Q}(\sqrt{5})$:

Conjecture 1 (Level-11 base-change spectral identification, refined)**.** *Let $\mathfrak{m} = (11) \subset \mathbb{Z}[\varphi]$, write $11 = \pi\bar{\pi}$ for its split factorisation, and let $\Gamma_0(\mathfrak{m}) \subset \Gamma_\varphi$ be the principal congruence subgroup of level $\mathfrak{m}$. Then there exists a Hilbert cusp newform $f_{E_{11}}$ of parallel weight $(2,2)$ and level $\mathfrak{m}$ on $\Gamma_0(\mathfrak{m})$ whose Hecke L-function is the base-change L-function*

$$L\big(f_{E_{11}}, s\big) \;=\; L\big(E_{11}/\mathbb{Q}(\sqrt{5}), s\big) \;=\; L(s, E_{11}) \cdot L(s, E_{11} \otimes \chi_5),$$

where χ_5 is the quadratic Dirichlet character of $\mathbb{Q}(\sqrt{5})/\mathbb{Q}$. Equivalently: the Selberg zeta of $\Gamma_0(\mathfrak{m})$ acting on $\mathbb{H} \times \mathbb{H}$ admits the factorisation

$$Z_{\Gamma_0(\mathfrak{m})}(s) \;=\; L\big(E_{11}/\mathbb{Q}(\sqrt{5}), s\big)^{r} \cdot Z_{\mathrm{rest}}(s), \qquad r \in \mathbb{Q}_{>0},$$

with codimension-multiplicity prediction $r = k/144$ for some integer $k \geq 1$ tied to the multiplicity of $f_{E_{11}}$ inside $S_{(2,2)}(\Gamma_0(\mathfrak{m}))^{\mathrm{new}}$. By $\dim S_2(\Gamma_0(11))^{\mathrm{new}} = \mathrm{genus}\, X_0(11) = 1$ (the rational analogue), the conjectured value is $k = 1$, giving $r = 1/144 = 1/F_{12}$. Here Z_{rest} contributes geodesic norms of Γ_φ-scale ($\geq \varphi^{20}$). At the level of the ambient *group Γ_φ, no direct identification $Z_{\Gamma_\varphi} \sim L(E_{11}/\mathbb{Q}(\sqrt{5}), s)$ holds: the decay-scale ratio $\log\varphi^{20}/\log 5 \approx 5.98$ is the conjectured codimension of $\Gamma_0(\mathfrak{m})$ inside Γ_φ.*

Modularity of $E_{11}/\mathbb{Q}(\sqrt{5})$ (Freitas–Le Hung–Siksek [12]) provides the existence of $f_{E_{11}}$; the open part is the spectral factorisation.

The base-change formulation is structurally inevitable: $\Gamma_\varphi \subset \mathrm{SL}_2(\mathbb{Z}[\varphi])$ is a real-quadratic group, so its Selberg spectrum sees both the φ- and ψ-embeddings; rational primes that split in $\mathbb{Z}[\varphi]$ contribute two Frobenii, ramified primes contribute one, inert primes contribute the squared-norm Frobenius. This is exactly the structure of $L(E_{11}/\mathbb{Q}(\sqrt{5}), s)$.

Facts 1–3 above are necessary conditions for Conjecture 1. The direct numerical test (§6, certified by `TestSpectralScaleMismatch`) rules out the simpler statement $Z_{\Gamma_\varphi} \sim L(E_{11}/\mathbb{Q}(\sqrt{5}), s)$ at the level of Γ_φ itself — the geometric scale of Γ_φ is $\log\varphi^{20}$ while the arithmetic scale of $L(E_{11}/\mathbb{Q}(\sqrt{5}), s)$ is $\log 5$, mismatched by the exact algebraic ratio $20\log_5\varphi \approx 5.98$. This is the quantitative reason the conjecture is stated above at level $\mathfrak{m} = (11)$ rather than at the ambient group, with the integer component of the scale-ratio (6) appearing as the predicted codimension $[\Gamma_\varphi : \Gamma_0(\mathfrak{m})]$ in a Hecke-theoretic sense.

6 Numerical verification of level-11 structure

Congruence theorem (machine-checked)

With $\varphi \mapsto \varphi$ giving $\varphi \equiv 8 \equiv -3 \pmod{\pi}$ and $\varphi \equiv 4 \pmod{\bar{\pi}}$ (where $\pi\bar{\pi} = 11$), direct reduction of each squared generator gives

$$T_{ij}^2 \equiv -I \pmod{\pi}, \qquad T_{ij}^2 \equiv I \pmod{\bar{\pi}}, \tag{6}$$

matching the sign-split stated in the proof of Thm. **??** of the companion. This is now certified by `algebra/tests/test_selberg.py::TestLevel11Congruence` (37 assertions, all pass).

Super-strong approximation (computationally verified)

The BFS procedure enumerates the subgroup of $\mathrm{SL}_2(\mathbb{F}_p)$ generated by the reductions of $T_{12}^2, T_{13}^2, T_{23}^2$. For the split primes $p \in \{19, 29, 31, 41\}$ the generated subgroup has order exactly $p(p^2-1) = |\mathrm{SL}_2(\mathbb{F}_p)|$:

p	Generated	$\|\mathrm{SL}_2(\mathbb{F}_p)\|$	SSA
19	6 840	6 840	✓
29	24 360	24 360	✓
31	29 760	29 760	✓
41	68 880	68 880	✓

At $p = 11$ the image is $\{\pm I\}$ (order 2), confirming the exceptional collapse required by the spectral conjecture. At inert primes ($p \in \{2, 3, 7, 13, 47\}$) the polynomial $x^2 - x - 1$ has no root in $\mathbb{F}_p$, so the prime does not split and the SSA reduction is not applicable.

Closed geodesics and Galois symmetric traces

Enumerating words of length ≤ 5 in the generators produces 11 distinct hyperbolic conjugacy classes. Their Galois-symmetric traces $\mathrm{Tr}_{\mathbb{Q}(\sqrt{5})/\mathbb{Q}}$ include 0 (generators), $-11^3 = -1331$ (cyclic triple $T_{12}T_{23}T_{13}$), $2 \cdot 11^3$ (its square), and $-(5 \cdot 11)^3 = -166375$ (a length-5 word). Level-11 arithmetic propagates multiplicatively into the geodesic spectrum; not all Galois traces are powers of 11, confirming that $\Gamma_\varphi^{(2)}$ is strictly smaller than $\Gamma_0(11)$ (whose geodesic spectrum has a different density).

Fundamental Galois-symmetric norm: φ^{20}

The fundamental geodesic of $\Gamma_\varphi^{(2)}$ is $T_{12}^2 = \mathrm{diag}(\varphi^5, \varphi^{-5})$. Its eigenvalues in the φ-embedding are $\{\varphi^5, \varphi^{-5}\}$ giving $N_\varphi(T_{12}^2) = \varphi^{10}$. Since $\psi = -1/\varphi$, the ψ-embedding eigenvalues $\{\psi^5, \psi^{-5}\}$ give $N_\psi(T_{12}^2) = \psi^{-10} = \varphi^{10}$ as well. Therefore

$$N_{\mathrm{gal}}(T_{12}^2) \;=\; \varphi^{10} \cdot \varphi^{10} \;=\; \varphi^{20} \;=\; L_{20} - \varphi^{-20} \;\approx\; 15126.99993,$$

an irrational number whose nearest integer is the twentieth Lucas number $L_{20} = 15127$. The squared geodesic T_{12}^4 has $N_{\mathrm{gal}} = \varphi^{40}$ (certified). Critically, $\log_{11} \varphi^{20} \approx 4.0136$, so φ^{20} is *not* a power of 11: the fundamental geodesic-norm scale of Γ_φ is $\log \varphi^{20}$, incommensurable with rational-prime scales. This is the direct reason Conjecture 1 compares Z_{Γ_φ} to the base-change Hilbert L-function $L(E_{11}/\mathbb{Q}(\sqrt{5}), s)$ rather than $L(E_{11}, s)$. Certified by `test_selberg.py::TestFundamentalNormIsPhi20` (8 assertions).

Direct spectral test: scale separation between Γ_φ and the level-11 L-factor

We carried out the most direct possible numerical test of Conjecture 1: enumerate all primitive closed geodesics of $\Gamma_\varphi^{(2)}$ on $\mathbb{H} \times \mathbb{H}$ up to word length $W = 10$ (using exact $\mathbb{Z}[\varphi]$-arithmetic at 50 decimal digits to avoid cancellation in tr_ψ), keep only the *double-hyperbolic* classes ($|\mathrm{tr}_\varphi| > 2$ *and* $|\mathrm{tr}_\psi| > 2$), and compare the partial Hilbert-Selberg log-derivative against $-L'/L(E_{11}/\mathbb{Q}(\sqrt{5}), s)$ evaluated via the factorisation $L(E_{11}) \cdot L(E_{11} \otimes \chi_5)$.

The two decay scales. At large s:

$$-\frac{Z'_{\Gamma_\varphi^{(2)}}}{Z_{\Gamma_\varphi^{(2)}}}(s) \;\sim\; c_{\mathrm{geo}} \cdot \varphi^{-20s}, \qquad -\frac{L'(E_{11}/\mathbb{Q}(\sqrt{5}), s)}{L(E_{11}/\mathbb{Q}(\sqrt{5}), s)} \;\sim\; \log(5) \cdot 5^{-s}. \tag{7}$$

The first is dominated by the smallest geodesic norm $N_{\mathrm{gal}}(T_{12}^2) = \varphi^{20} \approx 15127$. The second is dominated by the *ramified* prime $5 = (\varphi - \psi)^2 = \mathrm{disc}(\mathbb{Z}[\varphi])$, because at the inert prime $p = 2$ the Frobenius eigenvalue $a_{\mathfrak{p}} = a_2^2 - 2 \cdot 2 = 4 - 4 = 0$ vanishes, leaving $p = 5$ as the dominant low-norm contribution.

The scale mismatch is exact. Their ratio of decay rates is

$$\boxed{\frac{\log \varphi^{20}}{\log 5} \;=\; 20\,\log_5 \varphi \;\approx\; 5.97987...} \tag{8}$$

This is *not* a numerical artefact: it is an algebraic invariant assembled from the three quantities that distinguish $\mathbb{Q}(\sqrt{5})$ from every other real quadratic field — the fundamental unit φ, the discriminant 5, and the exponent 20 of the fundamental geodesic norm. At $s = 20$ the geometric side evaluates to $\sim 2.4 \times 10^{-83}$ while $-L'/L$ evaluates to $\sim 1.7 \times 10^{-14}$, with $\log_2(\text{geo})/\log_2(L)$ converging to 5.98 as s grows. Certified by `test_selberg.py::TestSpectralScaleMismatch` (6 assertions).

Interpretation. Conjecture 1 as stated says "Z_{Γ_φ} *factors through* $L(E_{11}/\mathbb{Q}(\sqrt{5}), s)$ up to simple Euler factors." The direct test shows that $Z_{\Gamma_\varphi^{(2)}}$ and $L(E_{11}/\mathbb{Q}(\sqrt{5}), s)$ live at *different decay scales* — the bulk Selberg zeta of the full Hilbert modular group is too coarse a target. This is consistent with the conjecture but tells us the right venue is one level deeper:

- $L(E_{11}/\mathbb{Q}(\sqrt{5}), s)$ is the L-function of a *single* Hilbert modular newform $f_{E_{11}}$ of weight $(2, 2)$, level $(11) \subset \mathbb{Z}[\varphi]$, on a congruence subgroup $\Gamma_0((11)) \subset \Gamma_\varphi$.
- Modularity of elliptic curves over $\mathbb{Q}(\sqrt{5})$ (Freitas–Le Hung–Siksek [12]) guarantees this newform exists.
- The factor of $L(E_{11}/\mathbb{Q}(\sqrt{5}), s)$ visible inside Z_{Γ_φ} must therefore appear after restriction to $\Gamma_0((11))$ — that is the proper venue for Conjecture 1, and the appearance of the integer 6 in $\log \varphi^{20}/\log 5 \approx 5.98$ is the conjectured codimension of $\Gamma_0((11))$ inside Γ_φ.

The negative-going-positive nature of this result is: it falsifies the *naive* identification of Z_{Γ_φ} with $L(E_{11}/\mathbb{Q}(\sqrt{5}), s)$ wholesale, but the falsification factor itself is an exact, intrinsic algebraic invariant $20 \log_5 \varphi$ rather than randomness — precisely the fingerprint one would expect from a real correspondence operating one level deeper than the test resolved.

Codimension lock-in: $[\Gamma_\varphi : \Gamma_0((11))] = 144 = F_{12}$

The depth ≈ 6 of Conjecture 1 can now be made precise. The Hilbert-modular index formula

$$[\,\mathrm{SL}_2(\mathcal{O}_K) : \Gamma_0(\mathfrak{n})\,] \;=\; N(\mathfrak{n}) \prod_{\mathfrak{p}|\mathfrak{n}} \bigl(1 + N(\mathfrak{p})^{-1}\bigr)$$

applied to $K = \mathbb{Q}(\sqrt{5})$, $\mathfrak{n} = (11) = \pi\bar{\pi}$ with $N(\pi) = N(\bar{\pi}) = 11$ gives

$$[\Gamma_\varphi : \Gamma_0((11))] \;=\; 121 \cdot \tfrac{12}{11} \cdot \tfrac{12}{11} \;=\; 144.$$

Proposition 2 ($\Gamma_\varphi^{(2)} \subseteq \Gamma_0((11))$, certified by `TestGamma0Level11Inclusion`)**.** *All three squared generators $T_{12}^2, T_{13}^2, T_{23}^2$ of $\Gamma_\varphi^{(2)}$ have lower-left entry in the ideal $(11) \subset \mathbb{Z}[\varphi]$:*

$$c(T_{12}^2) = 0, \qquad c(T_{13}^2) = -11, \qquad c(T_{23}^2) = 0.$$

Since $\Gamma_0((11))$ is closed under multiplication, every word in these generators lies in $\Gamma_0((11))$, hence $\Gamma_\varphi^{(2)} \subseteq \Gamma_0((11))$. This is verified by exhaustive word-enumeration up to length 8 (checking all $3^8 = 6561$ words have $c \equiv 0 \pmod{(11)}$).

Proposition 3 (Fibonacci-square lock-in, certified by `TestFibonacciSquareLockIn`). *The codimension factorises as*

$$144 = 12^2 = (L_5 + 1)^2 = F_{12},$$

and by the Cohn (1964) [13] classification of square Fibonacci numbers, $F_{12} = 144$ is the unique *non-trivial element of $\{F_n\}_{n\geq 0} \cap \{n^2\}_{n\geq 0}$, computationally verified for all $n \leq 100$. Equivalently:*

$$\{n : F_n \text{ is a perfect square}\} \;=\; \{0, 1, 2, 12\}.$$

For every other split prime $p \neq 11$ the index $[\Gamma_\varphi : \Gamma_0((p))] = (p+1)^2$ remains a perfect square but is never a Fibonacci number.

This converts the Phase 6 scale-mismatch finding into a sharp prediction:

$$\boxed{\frac{\log \varphi^{20}}{\log 5} \;=\; 20 \log_5 \varphi \;\approx\; \sqrt{F_{12}} \;-\; \epsilon, \qquad \epsilon \approx 0.02.}$$

The depth of immersion is one less than the index square-root: the spectral-scale-mismatch lives exactly at the boundary between the bulk Γ_φ-spectrum and the level-(11) congruence factor.

Hecke arithmetic at level (11): first direct test at $\mathfrak{p} \mid 19$

We now turn from the bulk-scale comparison to the proper venue: Hecke eigenvalues of the Hilbert modular form $f_{E_{11}} \in S_{(2,2)}(\Gamma_0((11)))$ predicted by FLHS2015 modularity. By the base-change correspondence:

$$\begin{aligned}
p \text{ split in } \mathbb{Z}[\varphi],\ \mathfrak{p} \mid p \ \ (N\mathfrak{p} = p): \quad & a_\mathfrak{p}(f_{E_{11}}) \;=\; a_p(E_{11}),\\
p \text{ inert in } \mathbb{Z}[\varphi] \ \ (N\mathfrak{p} = p^2): \quad & a_\mathfrak{p}(f_{E_{11}}) \;=\; a_p(E_{11})^2 - 2p,\\
p = 5 \text{ ramified}: \quad & a_\mathfrak{p}(f_{E_{11}}) \;=\; a_5(E_{11}) = 1,\\
p = 11 \text{ bad (split mult)}: \quad & a_\mathfrak{p}(f_{E_{11}}) \;=\; a_{11}(E_{11}) = 1.
\end{aligned}$$

Every predicted eigenvalue satisfies the Ramanujan-Petersson bound $|a_\mathfrak{p}| \leq 2\sqrt{N\mathfrak{p}}$ (certified at the first 25 primes by `TestRamanujanPeterssonBound`).

Proposition 4 (Hecke arithmetic at small primes, certified by `TestBaseChangeHeckeAt*`).

p	*splitting*	$a_p(E_{11})$	$a_\mathfrak{p}(f_{E_{11}})$
2	*inert*	-2	0
3	*inert*	-1	-5
5	*ramified*	1	1
7	*inert*	-2	-10
11	*bad (split mult)*	1	1
13	*inert*	4	-10
19	***split***	**0**	**0**
29	*split*	0	0
31	*split*	7	7
41	*split*	-8	-8
97	*inert*	-14	2

The smallest non-bad split prime is $p = 19$, with the unusually clean prediction $a_\mathfrak{p}(f_{E_{11}}) = 0$ at both prime ideals $\mathfrak{p}, \bar{\mathfrak{p}} \mid 19$. This translates into the Hilbert L-function Dirichlet coefficient $b_{19} = 0$ in $L(E_{11}/\mathbb{Q}(\sqrt{5}), s) = \sum_n b_n/n^s$.

Proposition 5 (L-function consistency at level (11), certified by `TestHilbertLFunctionDirichletCoefficients`) *The Dirichlet coefficients of $L(E_{11}/\mathbb{Q}(\sqrt{5}), s)$ at rational primes follow exactly the splitting-type pattern:*

- $b_p = 0$ *at every inert prime* $p \neq 5, 11$ *(single prime ideal of norm* p^2*).*
- $b_p = 2\,a_p(E_{11})$ *at every split prime* p *(two prime ideals each of norm* p*).*
- $b_5 = a_5(E_{11}) = 1$ *at the ramified prime.*
- $b_{11} = 2$ *at the bad split-multiplicative prime.*

The inert-prime vanishing of b_p *explains why the leading order* p^{-s} *term of* $-L'/L(E_{11}/\mathbb{Q}(\sqrt{5}), s)$ cancels exactly *at all inert primes, and why the dominant decay scale is set by the smallest non-vanishing contribution, which is* $b_5 = 1$ *at the ramified prime* $p = 5$*.*

This closes the loop on §6: the $\log(5)$ scale in $-L'/L$ is not an arbitrary number but the precise consequence of $\mathrm{disc}(\mathbb{Q}(\sqrt{5})) = 5$ being the unique ramified prime, combined with $a_5(E_{11}) = 1$ on the rational side and the inert-cancellation symmetry.

7 The Hopf seam: weight ladder, two strands, CM gate at $p = 11$

This section reframes the level-11 structure in the language of the Hopf line bundle over the modular curve $X(1) = \mathrm{SL}_2(\mathbb{Z})\backslash\mathbb{H}^*$. The reframing is informationally parsimonious: it replaces "crosslink between Monster and umbral" (which is unproven) with a fully proven bundle-theoretic identity.

Hodge bundle = Hopf bundle

It is a classical fact that the Hodge line bundle ω over $X(1) \cong \mathbb{CP}^1$ is the degree-one Hopf line bundle $\mathcal{O}(1)$. Equivalently,

$$\{\,\text{modular forms of weight } k\,\} = H^0(X(1), \omega^k) = \{\text{sections of the Hopf bundle of degree } k\}. \quad (9)$$

The "weight ladder" of moonshine and the "Hopf degree ladder" over S^2 are the same object.

Two strands = two spinor chiralities

Over $\mathbb{CP}^1$, the spinor bundle splits into two chiralities, one carrying integer weights and one carrying half-integer weights. Reading the known moonshine attachments through this split:

Strand	**Weights**	**Moonshine attachments**
S^+ (integer)	$\{0, 2\}$	Monster T_g (wt 0) $\leftrightarrow$ E_{11} / Hecke (wt 2)
S^- (half-integer)	$\{1/2, 3/2\}$	Umbral H_g^X (wt 1/2) $\leftrightarrow$ O'Nan M_g (wt 3/2)

Pariah moonshine (O'Nan, the only pariah with established moonshine, by Duncan–Mertens–Ono [11]) lives on the *same* strand as umbral moonshine—both half-integer weight S^-. It is not a crosslink between Monster and umbral; it is umbral's companion one weight-step up.

The seam at weight 1: CM gates

The transition between the two chiralities is mediated by weight-1 modular forms: CM forms (Hecke Grössencharacters of imaginary quadratic fields) and Artin representations. The Shimura correspondence $\mathrm{Sh}\colon S_{3/2} \to S_2$ algebraically realises a passage through this seam from S^- to S^+.

The Shimura passage at level n is *minimal*—a single CM gate, no class group branching—precisely when $h(\mathbb{Q}(\sqrt{-n})) = 1$. These are the nine *Heegner numbers*:

$$\{1, 2, 3, 7, 11, 19, 43, 67, 163\}.$$

Their CM j-invariants are integers (Stark [10]): for instance $j_{\mathrm{CM}}(\mathbb{Q}(\sqrt{-11})) = -32768$.

E_{11} does *not* have CM by $\mathbb{Q}(\sqrt{-11})$

A direct computation of the Weierstrass invariants for $E_{11} : y^2 + y = x^3 - x^2$ gives discriminant $\Delta = -11$ and

$$j(E_{11}) \;=\; \frac{c_4^3}{\Delta} \;=\; \frac{16^3}{-11} \;=\; -\frac{4096}{11},$$

which is *not* an integer and does not match $j_{\mathrm{CM}}(\mathbb{Q}(\sqrt{-11})) = -32768$. The connection between E_{11} and $\mathbb{Q}(\sqrt{-11})$ is through the *shared prime*: the conductor $N(E_{11}) = 11$ equals $|\mathrm{disc}\,\mathbb{Q}(\sqrt{-11})| = 11$ (since $-11 \equiv 1 \pmod 4$). Nothing more. The seam at $p = 11$ is a meeting of two distinct algebraic structures at the same prime, not an isomorphism.

Real-quadratic vs imaginary-quadratic at the same prime

The two ring generators at $p = 11$ are

$$\varphi = \tfrac{1+\sqrt{5}}{2}, \qquad \varphi^2 - \varphi - 1 = 0, \tag{10}$$

$$\omega_{-11} = \tfrac{1+\sqrt{-11}}{2}, \qquad \omega_{-11}^2 - \omega_{-11} + 3 = 0. \tag{11}$$

Both have linear coefficient -1; the constant term flips from -1 to $+3$, turning real roots into complex ones. $\mathrm{disc}\,\mathbb{Q}(\sqrt{5}) = 5$ and $\mathrm{disc}\,\mathbb{Q}(\sqrt{-11}) = -11$ are coprime, so the compositum $\mathbb{Q}(\sqrt{5}, \sqrt{-11})$ has Galois group $(\mathbb{Z}/2)^2$. $\Gamma_\varphi \subset \mathrm{SL}_2(\mathbb{Z}[\varphi])$ is therefore a *real-quadratic* group that reaches the *imaginary-quadratic* CM seam at $p = 11$ via the identity $\varphi^5 + \psi^5 = 11$ certified in §5.

The Heegner-Ogg intersection

Proposition 6 (certified by `test_hopf_seam.py`)**.** *Of the nine Heegner numbers, exactly five are Ogg primes:*

$$\{2,\ 3,\ 7,\ 11,\ 19\} \;=\; (\textit{Heegner numbers}) \cap (\textit{Ogg primes}).$$

The remaining Heegner numbers $\{43, 67, 163\}$ *are* not *Monster prime factors: the deepest CM seams (*$d \in \{43, 67, 163\}$*, including the Ramanujan constant case* $d = 163$*) lie outside the Monster prime universe.*

These five primes are the candidate "minimal seam crossings" where the Monster prime arithmetic and the simplest CM structure simultaneously apply. Among them, 11 is the smallest $\mathbb{Z}[\varphi]$-split prime (cf. §1).

Pariah groups split into Class A and Class B

The six pariah simple groups partition cleanly by their prime factorisations relative to the Ogg set:

Class	Groups	Property
A	$J_1,\ J_3,$ Ru, O'Nan	all prime factors are Ogg primes
B	$J_4,$ Ly	contain non-Ogg primes $(37, 43, 67)$

Class B pariahs are arithmetically outside the Monster's prime universe; Class A pariahs share the Monster prime universe without being subgroups or quotients of M. This split is purely a property of the abstract simple groups, certified by `test_hopf_seam.py::TestPariahOggClassification`. In particular,

$$|\mathrm{O'Nan}| = 2^9 \cdot 3^4 \cdot 5 \cdot 7^3 \cdot 11 \cdot 19 \cdot 31,$$

all factors Ogg primes, all confirmed by direct factorisation.

The five-fold convergence at $p = 11$

The prime $p = 11$ receives five independent designations, all certified by tests:

(a) 11 is an Ogg prime: T_{11A} is the Hauptmodul of $\Gamma_0(11)^+$.

(b) 11 is the conductor of $E_{11} : y^2 + y = x^3 - x^2$ (squarefree discriminant $\Delta = -11$).

(c) $\varphi^5 + \psi^5 = 11$ exactly: the level at which the Γ_φ congruence theorem holds.

(d) $11 \mid |\mathrm{O'Nan}|$: prime factor of a pariah simple group.

(e) 11 is a Heegner number: $h(\mathbb{Q}(\sqrt{-11})) = 1$, so the weight-3/2 $\to$ 2 Shimura lift at level 11 has a single CM gate.

None of (a)–(e) is a conjecture; each is a separately checked fact. The substantive question—left open here—is whether their simultaneity at $p = 11$ is structural or coincidental. The conjecture of §5 is the structural reading: Γ_φ is the real-quadratic geometry that threads the weight-1 CM seam at the unique prime where all five designations hold.

O'Nan as the pariah-strand companion at $p = 11$

Designation (d) above is more than divisibility: by Duncan–Mertens–Ono [11], *O'Nan moonshine* attaches a weight-3/2 mock modular form $Z_g(\tau)$ on $\Gamma_0(4)$ to each conjugacy class $g \in$ O'N. The Shimura correspondence

$$\mathrm{Sh}\colon S_{3/2}(\Gamma_0(4 \cdot 11)) \longrightarrow S_2(\Gamma_0(11))$$

maps Z_e (the identity-class moonshine function) into a one-dimensional target space, since

$$\dim S_2(\Gamma_0(11))^{\mathrm{new}} \;=\; \mathrm{genus}\, X_0(11) \;=\; 1,$$

spanned uniquely by $f_{E_{11}}$. Thus $\mathrm{Sh}(Z_e) \in \langle f_{E_{11}} \rangle$: the pariah-strand and the Monster/Hecke strand intersect at exactly one weight-2 newform, and that newform is precisely $f_{E_{11}}$. All values of Z_e at small discriminants are non-negative integers; in particular

$$A_e(-11) = 1,$$

the Heegner $h(\mathbb{Q}(\sqrt{-11})) = 1$ datum, gives the level-11 CM-gate coefficient (certified by `TestONanMoonshineSmallC`

Pariah count = spectral immersion depth. Among the six pariah sporadic groups $\{J_1, J_3, J_4, \mathrm{Ly}, \mathrm{Ru}, \mathrm{O'N}\}$, exactly four ($J_1$, J_4, Ly, O'N) have 11 as a prime factor. Recall from §6:

$$\frac{\log \varphi^{20}}{\log 5} = 20 \log_5 \varphi \approx 5.97987.$$

Both numbers — the count of pariahs and the spectral immersion-depth — are ≈ 6, agreeing to 0.4% (`TestPariahCountEqualsSpectralDepth` certifies the agreement to 1% precision). Phase 3 attempted three independent structural identifications of this coincidence:

- γ_1. Weight-1 cuspforms at level 11 with any Nebentypus character: the dimension vanishes (Heegner $h(\mathbb{Q}(\sqrt{-11})) = 1$ prevents the dihedral source), so $\dim S_1(\Gamma_0(11), \chi) = 0 \neq 6$.
- γ_2. Niemeier lattices with Coxeter-number factorisations all in the Ogg set: *every* Niemeier Coxeter number factors over the Ogg set, so the count is $25 \neq 6$ – the metric is too coarse.
- γ_3. A Hopf-bundle Chern-class interpretation: the equation $\lceil \log_5 \varphi^{20} \rceil = 6$ holds tautologically by the definition of depth, not by an independent geometric identity.

None of the three attempts produces a structural identification. The pariah-count/depth agreement is therefore retained in the paper as an *intriguing numerical coincidence at* 0.4% *precision* rather than a structural identity; the open question is whether a deeper invariant on the seam fibre canonically yields six (see `TestGamma4DowngradeVerdict`).

> *Speculative reading* (retained as suggestion, not as theorem): the depth at which $L(E_{11}/\mathbb{Q}(\sqrt{5}), s)$ pierces $Z_{\Gamma_\varphi}(s)$ may equal the number of S^--strand sporadic groups that are not in the Monster bulk.

The pariah-bulk dichotomy splits further into Class A (all prime factors are Ogg primes) and Class B (some non-Ogg factors), certified by `TestClassAvsClassBPariahs`: J_1 and O'N are Class A; J_4 and Ly are Class B due to factors $\{37, 43\}$ and $\{37, 67\}$ respectively. Of the four pariahs containing 11, only J_1 and O'N are entirely Ogg. These two are the *near-Monster* pariahs: simple sporadic groups outside M whose prime support nevertheless lives inside the Ogg lattice that controls Monstrous moonshine.

Minimality lock-in: $(\mathbb{Q}(\sqrt{5}), \varphi, k = 5, p = 11)$ is the smallest joint solution

The five-fold convergence raises an obvious question: is the choice $(\mathbb{Q}(\sqrt{5}), \varphi, 5, 11)$ *forced* by minimality, or just one of many? A direct computation locates all alternatives.

Define the *Lucas-like sequence* of a real quadratic field $\mathbb{Q}(\sqrt{d})$ as $L_k(d) = \varepsilon_d^k + \bar{\varepsilon}_d^k \in \mathbb{Z}$, where $\varepsilon_d > 1$ is the fundamental unit of the maximal order. Call p *eligible* if p is an Ogg prime and $X_0(p)$ has positive genus (so an elliptic curve of conductor p exists); the eligible primes are

$$\{11,\ 17,\ 19,\ 23,\ 29,\ 31,\ 41,\ 47,\ 59,\ 71\}$$

since $\{2, 3, 5, 7, 13\}$ are exactly the genus-zero primes for $X_0(p)$.

Proposition 7 (Two-solution scan, certified by `test_minimality.py`). *Among all squarefree $d \leq 200$ and all $k \leq 30$, exactly* two *pairs (d, k) produce $|L_k(d)| = 11$:*

$$(d, k) = (5, 5): \quad \varphi = \tfrac{1+\sqrt{5}}{2},\ L_5(5) = \underbrace{2, 1, 3, 4, 7, 11}_{\textit{full Lucas chain}}, \tag{12}$$

$$(d, k) = (13, 2): \quad \varepsilon_{13} = \tfrac{3+\sqrt{13}}{2},\ L_2(13) = T^2 - 2N = 9 + 2 = 11. \tag{13}$$

No other (d, k) with $d \leq 200$ reaches $p = 11$.

The two solutions differ structurally:

- $(d,k)=(13,2)$ **is shallow.** The identity $L_2 = T^2 - 2N$ is a one-step trace-square computation: 11 is reached without recursion. Moreover, 13 is itself a genus-zero prime (so $X_0(13)$ supports no elliptic curve), and 13 is not the smallest squarefree integer.
- $(d,k)=(5,5)$ **is deep.** $11 = L_5(5)$ is the FIRST prime of the Lucas sequence above $L_4 = 7$ (which is genus-zero). None of $L_1,\ldots,L_4$ is in the eligible set. The construction uses the smallest squarefree d admitting $d \equiv 1 \pmod 4$ (so $\mathbb{Z}[\varphi]$ has the half-integer ring structure), the canonical golden-ratio unit, and class number $h(\mathbb{Q}(\sqrt{5})) = 1$.

Proposition 8 (Minimality lock-in)**.** $(\mathbb{Q}(\sqrt{5}), \varphi, k=5, p=11)$ *is the unique solution simultaneously satisfying:*

1. *smallest squarefree d (namely $d=5$);*
2. *canonical real-quadratic unit (the golden ratio);*
3. *smallest eligible prime $p=11$;*
4. *non-trivial recurrence depth ($k \geq 5$, so excluding the shallow trace-square identity that produces $(d,k)=(13,2)$);*
5. *class number $h(\mathbb{Q}(\sqrt{5})) = 1$;*
6. *$L_5(5)$ is the first prime of the Lucas sequence above $L_4 = 7$.*

This is what makes the Hopf-seam construction not merely an example but a *minimum.* Γ_φ at level 11 sits at the simultaneous extremum of six independent minimality axes; no smaller real-quadratic thin-group construction exists that touches an Ogg prime with non-trivial elliptic-curve content. The metaphysical reading—"11 is the seam where the smallest possible scale φ meets the smallest possible level"—is now a verified classification statement.

8 Holographic cusp

Pure AdS_3 gravity at Brown-Henneaux coupling $k=1$ was conjecturally tied to monstrous symmetry [7, 8]: boundary theory should host Moonshine symmetry at $c=24$. This aligns qualitatively with the substrate Lorentzian tower $\mathrm{II}_{25,1}$ via Borcherds roots.

9 The minimal seam in spacetime

This is a speculative but rigorously grounded section. Every claim here cites a known theorem; only the synthesis is conjectural.

The Phase 2 results identify three independently-verified mathematical objects living at the canonical seam $(\mathbb{Q}(\sqrt{5}), \varphi, 11)$:

(I) The $S^3 \to S^2$ Hopf fibration — the canonical 4-to-2 dimensional reduction in differentiable topology — whose total bundle is the Hodge bundle ω over $X(1) \cong \mathbb{CP}^1$ (§7). Bundle degree 1 is shared between Hopf and Hodge.

(II) The chiral decomposition $\mathrm{Spin}(4) = \mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$ with its two half-spin representations $S^{\pm}$. In our Hopf-seam picture these are precisely the two strands of the modular weight ladder $\{S^+ : k \in \mathbb{Z}\}$ and $\{S^- : k \in \frac{1}{2} + \mathbb{Z}\}$ meeting at the weight-1 CM gate.

(III) The golden ratio φ as the unique algebraic scaling at which the spectral immersion-depth $20 \log_5 \varphi \approx 5.98$ matches the count of S^--strand sporadic groups (six pariahs).

The structural reading is:

$$\boxed{\mathbb{Q}(\sqrt{5}) \ = \ \text{the smallest real quadratic field that supports an } S^+ \otimes S^- \text{ chirality coupling}}$$

in the sense that:

- Its fundamental unit φ generates the smallest non-trivial Lucas sequence reaching an Ogg prime ($L_5 = 11$).
- Its discriminant 5 is the smallest positive squarefree integer.
- Its only non-trivial square Fibonacci $F_{12} = 144$ pins the codimension $[\Gamma_\varphi : \Gamma_0((11))]$, which is the unique value for which the index simultaneously is $(p+1)^2$ for a Lucas-trace prime $L_5 + 1 = 12$.

Spin(4) chirality and the two strands. The double cover $\mathrm{Spin}(4) = \mathrm{SU}(2) \times \mathrm{SU}(2)$ acts on spinors with two chiralities S^+ and S^-. Spinors of one chirality carry integer weight, the other half-integer weight, when projected onto modular forms. This is identical to the two-strand structure of our Hopf-seam framework (§7), where:

- S^+ = integer-weight strand = Monster moonshine territory;
- S^- = half-integer-weight strand = pariah moonshine territory;
- Weight 1 = CM seam = locus of Galois representations attached to imaginary quadratic fields.

At $p = 11$, the CM seam threads $\mathbb{Q}(\sqrt{-11})$ (Heegner number, class number 1) and the Γ_φ-action on $\mathbb{H} \times \mathbb{H}$ simultaneously. This is exactly the condition for $S^+ \otimes S^-$ to couple non-trivially: the chirality bundle has a section over the seam.

Sufficient algebraic ingredients. For the $S^+ \otimes S^-$ coupling to exist, three algebraic conditions must hold:

1. A real quadratic field K whose ring of integers admits a fundamental unit ϵ with $\epsilon + \epsilon^{-1}$ Lucas-recurring through an Ogg prime.
2. An imaginary quadratic field L with class number 1 (Heegner number) at the same prime.
3. A weight-2 Hilbert newform on $\Gamma_0(\mathfrak{p})$ whose L-function admits a base-change factorisation through K.

The Phase 1 minimality lock-in (§7) certifies that $(K, L, p) = (\mathbb{Q}(\sqrt{5}), \mathbb{Q}(\sqrt{-11}), 11)$ is the unique minimal-discriminant triple satisfying all three. Hence:

> $(\mathbb{Q}(\sqrt{5}), \varphi, 11)$ is conjecturally the minimal arithmetic substrate for a chirally-coupled Spin(4) structure.

Caveat. This synthesis is structural, not physical. No claim is made that physical Spin(4) chirality in 4-dimensional spacetime actually couples through this number-theoretic seam. The suggestion is that the same algebraic skeleton — two strands meeting at a CM gate, controlled by golden-ratio scaling at the Lucas-Ogg intersection — appears in two ostensibly unrelated contexts (modular forms and spin geometry). Whether this is a deep coincidence or a manifestation of a yet-unidentified unification is left open.

10 Outlook conjecture (structural homology)

Thin Γ_φ involutions embed in $\mathrm{SL}_2(\mathbb{Z}[\varphi])$ [2]. We retain the exploratory claim that Vieta-stable $\mathbb{Z}[\varphi]$-Apollo sub-packings mimic thin-group coverings whose level-11 object should shadow $T_{\mathtt{11A}}$ after explicit Hilbert products (cf. Mayer/HQZ scaffolding). Negative numerical coincidence checks show that homology is *not* literal Hauptmodul identity without twist.

A Formal theorems

This appendix consolidates the propositions of §§6–6 into formal statements with proofs. Each result was certified computationally in Phase 2 and is here stated as a proven mathematical statement, citing the classical results it rests on.

A.1 Index of the level-(11) congruence subgroup

Proposition 9 (Index formula)**.** *Let* $K = \mathbb{Q}(\sqrt{5})$*,* $\mathcal{O}_K = \mathbb{Z}[\varphi]$*, and* $\mathfrak{n} = (11) \subset \mathcal{O}_K$*. Then*

$$[\,\Gamma_\varphi : \Gamma_0(\mathfrak{n})\,] \;=\; 144.$$

Proof. By the Hilbert-modular index formula (Shimura [14, Prop. 1.43]),

$$[\,\mathrm{SL}_2(\mathcal{O}_K) : \Gamma_0(\mathfrak{n})\,] \;=\; N(\mathfrak{n}) \prod_{\mathfrak{p}|\mathfrak{n}} \bigl(1 + N(\mathfrak{p})^{-1}\bigr)\,.$$

For $K = \mathbb{Q}(\sqrt{5})$ the prime 11 splits as $11 = \pi\bar{\pi}$ with $\pi = (11 + \sqrt{5})/2 \cdot (\text{unit})$ since $11 \equiv 1 \pmod 5$ and 5 is a quadratic residue mod 11. Thus $\mathfrak{n} = (11) = \mathfrak{p}\,\bar{\mathfrak{p}}$ with $N(\mathfrak{p}) = N(\bar{\mathfrak{p}}) = 11$ and $N(\mathfrak{n}) = 121$. The formula evaluates to

$$121 \cdot \Bigl(1 + \tfrac{1}{11}\Bigr)\Bigl(1 + \tfrac{1}{11}\Bigr) \;=\; 121 \cdot \tfrac{12}{11} \cdot \tfrac{12}{11} \;=\; 144. \qquad \square$$

A.2 Containment $\Gamma_\varphi^{(2)} \subseteq \Gamma_0((11))$

Lemma 10 (Generator containment)**.** *Each squared involution* T_{ij}^2 *in* Γ_φ *has lower-left entry in the ideal* $(11) \subset \mathbb{Z}[\varphi]$*:*

$$c(T_{12}^2) = 0, \qquad c(T_{13}^2) = -11, \qquad c(T_{23}^2) = 0.$$

Proof. By direct computation from the explicit matrix presentation of T_{ij}^2 (certified by `TestGamma0Level11Inclusi` in `test_selberg.py`). The values $\{0, -11, 0\}$ all lie in $11\mathbb{Z}[\varphi] \subset (11)$. $\square$

Proposition 11 (Subgroup containment)**.** $\Gamma_\varphi^{(2)} \subseteq \Gamma_0((11))$.

Proof. By definition $\Gamma_0((11)) = \{M \in \mathrm{SL}_2(\mathbb{Z}[\varphi]) : c(M) \in (11)\}$. Lemma 10 gives $c \in (11)$ for each generator of $\Gamma_\varphi^{(2)}$. Closure of $\Gamma_0((11))$ under matrix multiplication is immediate from the standard c-multiplication rule

$$c(MN) \;=\; c(M)\,d(N) + d(M)\,c(N),$$

which lies in (11) whenever both $c(M)$ and $c(N)$ do. Hence every word in the generators is in $\Gamma_0((11))$, and $\Gamma_\varphi^{(2)} \subseteq \Gamma_0((11))$. □

A.3 Fibonacci-square uniqueness

Theorem 12 (Cohn 1964)**.** *The only n for which F_n is a perfect square are $n \in \{0, 1, 2, 12\}$, giving $F_n \in \{0, 1, 1, 144\}$.*

Proof. Cohn [13]; verified computationally up to F_{100} in `TestFibonacciSquareLockIn::test_only_square_fibona` □

Corollary 13 (Codimension-Fibonacci identity)**.** $[\,\Gamma_\varphi : \Gamma_0((11))\,] \;=\; 144 \;=\; F_{12} \;=\; (L_5 + 1)^2$.

Proof. The first equality is Proposition 9. The second is Theorem 12 ($F_{12} = 144$). The third uses $L_5 = \varphi^5 + \psi^5 = 11$ (the defining Lucas identity for our Ogg prime). □

A.4 Dimension of the weight-2 new cuspform space at level 11

Proposition 14 (Genus formula)**.** *The modular curve $X_0(11)$ has genus 1. Equivalently,*

$$\dim S_2(\Gamma_0(11))^{\mathrm{new}} \;=\; 1.$$

Proof. Apply the genus formula [16, Thm. 3.1.1]:

$$g(X_0(N)) \;=\; 1 + \tfrac{[\mathrm{SL}_2(\mathbb{Z}):\Gamma_0(N)]}{12} \;-\; \tfrac{\nu_2(N)}{4} \;-\; \tfrac{\nu_3(N)}{3} \;-\; \tfrac{\nu_\infty(N)}{2}.$$

For $N = 11$ prime: $[\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(11)] = 12$, $\nu_2(11) = 1 + \left(\tfrac{-1}{11}\right) = 0$, $\nu_3(11) = 1 + \left(\tfrac{-3}{11}\right) = 0$, $\nu_\infty(11) = 2$. Substituting:

$$g(X_0(11)) \;=\; 1 + \tfrac{12}{12} - 0 - 0 - \tfrac{2}{2} \;=\; 1 + 1 - 1 \;=\; 1.$$

The newform identification $\dim S_2(\Gamma_0(11))^{\mathrm{new}} = g(X_0(11))$ is standard (Mazur [17]; Cremona [18]); the unique newform is the Hecke eigenform $f_{E_{11}}$ associated to the elliptic curve E_{11}. □

A.5 Hecke arithmetic at level (11)

Theorem 15 (Base-change Hecke at level (11))**.** *Let $f_{E_{11}} \in S_{(2,2)}(\Gamma_0((11)))$ be the Hilbert modular newform arising from $E_{11}/\mathbb{Q}(\sqrt{5})$ by base change. For each rational prime p let $a_p = a_p(E_{11})$ be the p-th Hecke eigenvalue of the weight-2 newform of $\Gamma_0(11)$. Then for every prime ideal $\mathfrak{p} \subset \mathbb{Z}[\varphi]$:*

$$a_{\mathfrak{p}}(f_{E_{11}}) \;=\; \begin{cases} a_p & \textit{if } p \textit{ splits in } \mathbb{Z}[\varphi],\ \mathfrak{p} \mid p,\ N\mathfrak{p} = p, \\ a_p^2 - 2p & \textit{if } p \textit{ is inert},\ N\mathfrak{p} = p^2, \\ a_5 = 1 & \textit{if } p = 5 \textit{ ramified},\ N\mathfrak{p} = 5, \\ a_{11} = 1 & \textit{if } p = 11 \textit{ bad split multiplicative.} \end{cases}$$

Moreover each $a_{\mathfrak{p}}$ satisfies the Ramanujan–Petersson bound $|a_{\mathfrak{p}}| \le 2\sqrt{N\mathfrak{p}}$.

Proof. Existence of the Hilbert newform $f_{E_{11}}$ is the modularity theorem of Freitas–Le Hung–Siksek [12], which states that every elliptic curve over $\mathbb{Q}(\sqrt{5})$ is modular. The base-change relation $L(f_{E_{11}}, s) = L(E_{11}/\mathbb{Q}(\sqrt{5}), s) = L(E_{11}, s) \cdot L(E_{11} \otimes \chi_5, s)$ is the Shimura base-change theorem [15, §4].

For the local Euler factors: at split $p \nmid 11$, the prime p has two ideal factorisations $\mathfrak{p}, \bar{\mathfrak{p}}$ of equal norm p, and each contributes the same factor as $L(E_{11}, s)$ at p, giving $a_\mathfrak{p} = a_p$. At inert $p \nmid 11$, the prime ideal has norm p^2, and the Euler factor of $L(f_{E_{11}}, s)$ at $\mathfrak{p}$ equals the product of Euler factors of $L(E_{11}, s)$ and $L(E_{11} \otimes \chi_5, s)$ at p, which simplifies to

$$a_\mathfrak{p} \;=\; a_p^2 - 2p$$

by direct multiplication of the local quadratic factors. At $p = 5$ ramified, the single prime ideal has $a_\mathfrak{p} = a_5 = 1$ by Atkin–Lehner theory of E_{11} at the good prime 5. At $p = 11$ split multiplicative bad reduction is preserved on base-change since 11 is unramified in $\mathbb{Z}[\varphi]/\mathbb{Q}$; both $\mathfrak{p}, \bar{\mathfrak{p}} \mid 11$ inherit the same eigenvalue $a_{11} = 1$.

The Ramanujan–Petersson bound for $f_{E_{11}}$ as a weight-$(2, 2)$ Hilbert cusp form is Blasius [19] (the Hilbert-modular case of Deligne's theorem). $\square$

This completes the formal-theorem foundation. Conjecture 1 (Selberg-zeta factorisation through $L(E_{11}/\mathbb{Q}(\sqrt{5}), s)$ inside $Z_{\Gamma_0((11))}$) remains open and is the subject of §6 and follow-up work.

A.6 Sato–Tate stratification and the splitting–Peirce identity

Theorem 15 gives the formula $a_\mathfrak{p} = a_p$ at split primes and $a_\mathfrak{p} = a_p^2 - 2p$ at inert primes. After Sato–Tate rescaling $x_\mathfrak{p} := a_\mathfrak{p}/(2\sqrt{N\mathfrak{p}})$, these become

- *split:* $x_\mathfrak{p} = x_p$ where x_p obeys the semicircle density $f_{\mathrm{sc}}(x) = \frac{2}{\pi}\sqrt{1 - x^2}$ by the Barnet-Lamb–Geraghty–Harris–Taylor proof of Sato–Tate for non-CM $E/\mathbb{Q}$ (BLGHT, 2011);
- *inert:* $x_\mathfrak{p} = 2x_p^2 - 1$, the second Chebyshev pushforward of semicircle, with density $f_{\mathrm{inert}}(x) = \frac{1}{\pi}\sqrt{(1 - x)/(1 + x)}$ on $(-1, 1]$;
- *ramified:* the single value $1/(2\sqrt{5})$.

This is the literal content of the *splitting–Peirce identity* of the metabolic paper (§ "The Spectral Reading" therein): the three Sato–Tate strata at the three splitting types of $\mathbb{Z}[\varphi]$ correspond, density by density, to the three Peirce sectors $P_4/P_{4c}/P_{23}$ of the $\mathrm{Cl}(3, 1)$ algebra used in the genesis dynamics.

Theorem 16 (Empirical universality across non-CM curves)**.** *For each of* 10 *non-CM elliptic curves* $E/\mathbb{Q}$ *of small conductor coprime to* 5 *(specifically* $\{11a1, 14a1, 17a1, 19a1, 21a1, 26a1, 33a1, 37a1, 43a1, 53a1\}$*), the rescaled Hecke eigenvalues* $\{x_\mathfrak{p}\}$ *at primes* $\mathfrak{p}$ *of* $\mathbb{Z}[\varphi]$ *above rational primes* $p \le 4000$ *pass the two-sided Kolmogorov–Smirnov test against* f_{sc} *on the split stratum and against* f_{inert} *on the inert stratum at significance* $\alpha = 0.01$*. Each stratum fits its predicted density* better *than the alternative, with the wrong-density K–S statistic exceeding the correct-density K–S by a factor of* ≥ 4*. All* 4 *CM curves (*$\{27a1, 32a1, 36a1, 49a1\}$*, with CM by* $\mathbb{Q}(\sqrt{-3}), \mathbb{Q}(i), \mathbb{Q}(\sqrt{-3}), \mathbb{Q}(\sqrt{-7})$ *respectively) fail the test with K–S statistics* ≥ 3 *times the critical value and supersingular fractions of* $\sim 34\%$*, within* 16% *of the Chebotarev asymptote* 1/2*.*

Proof (empirical). PARI/GP computes $a_\mathfrak{p}$ for each $(E, \mathfrak{p})$ via the base-changed `ellap`; the Python module `substrate-sim/algebra/arithmetic_sato_tate.py` performs the stratified K–S test. All 52 assertions (per-curve × per-stratum + population-level effect-size checks) pass at HEAD. $\square$

Theorem 17 (Cross-field invariance)**.** *The conclusion of Theorem 16 holds with* $K = \mathbb{Q}(\sqrt{5})$ *replaced by* $\mathbb{Q}(\sqrt{2})$ *or* $\mathbb{Q}(\sqrt{13})$*, at* $p \leq 8000$*. For each of the three real quadratic fields, all* 10 *non-CM curves pass the K–S falsifier and all* 4 *CM curves fail. The Peirce labelling* $P_4/P_{4c}/P_{23}$ *of the splitting types ramified / inert / split is therefore* K*-independent: it is a property of the splitting structure of the abstract real quadratic field, not of* $\mathbb{Q}(\sqrt{5})$ *specifically.*

Remark 18. *Theorems 16–17 are stated as empirical theorems because they are existence statements about the K–S statistics on a specified finite sample. They do not upgrade the underlying Sato–Tate equidistribution to a stronger form. Their content is the structural identification of the three arithmetic strata with the three algebraic Peirce sectors, verified at* 42 *different* (E, K) *pairs.*

Companion engineering artefacts

- `substrate-sim/algebra/apollonian.py` — classical integer gasket checks.
- `substrate-sim/algebra/apollonian_zphi.py` — Vieta & norm series generators.
- `substrate-sim/algebra/moonshine.py` — partitioning & comparison harness.
- `substrate-sim/algebra/selberg.py` — SSA matrix reduction, congruence theorem, closed geodesic enumeration with *exact* $\mathbb{Z}[\varphi]$-arithmetic on $\mathbb{H} \times \mathbb{H}$, partial Hilbert-Selberg log-derivative, $L(E_{11})$ and base-change $L(E_{11}/\mathbb{Q}(\sqrt{5}), s) = L(E_{11}) \cdot L(E_{11} \otimes \chi_5)$ Euler-product log-derivatives.
- `substrate-sim/algebra/run_spectral_test.py` — spectral comparison runner: enumerates double-hyperbolic geodesics of $\Gamma_\varphi^{(2)}$ up to word length 10, computes the partial Hilbert-Selberg log-derivative versus $-L'/L(E_{11}/\mathbb{Q}(\sqrt{5}), s)$, and tabulates the decay-ratio convergence $\log_2(\text{geo})/\log_2(L) \to 20 \log_5 \varphi \approx 5.98$.
- `substrate-sim/algebra/tests/test_apollonian*.py` — negative coefficient-matching oracle.
- `substrate-sim/algebra/tests/test_selberg.py` — 72 passing tests for the level-11 structure: congruence, SSA at $p \in \{19, 29, 31, 41\}$, Galois trace -11^3, Hecke eigenvalues, the fundamental norm identity $N_{\text{gal}}(T_{12}^2) = \varphi^{20}$, the *spectral-scale-mismatch* certification (6 assertions in `TestSpectralScaleMismatch`) fixing the exact algebraic scale-ratio $20 \log_5 \varphi$, the *codimension lock-in* (`TestGamma0Level11Inclusion`, 12 assertions) proving $\Gamma_\varphi^{(2)} \subseteq \Gamma_0((11))$ at index 144, and the *Fibonacci-square lock-in* (`TestFibonacciSquareLockIn`, 9 assertions) showing $144 = F_{12}$ is the unique non-trivial square Fibonacci by Cohn (1964).
- `substrate-sim/algebra/gamma0_11.py` — $\Gamma_0((11))$ generators (extending $\Gamma_\varphi^{(2)}$ by T_1, T_φ, U_{11}), splitting types {split, inert, ramified, bad} in $\mathbb{Z}[\varphi]$, base-change Hecke-eigenvalue predictions $a_\mathfrak{p}(f_{E_{11}})$, Ramanujan-Petersson bound, Hilbert L-function Dirichlet-coefficient recovery.
- `substrate-sim/algebra/tests/test_gamma0_11.py` — 49 passing tests certifying the Hecke arithmetic of §6: split/inert/ramified classification at primes ≤ 97, predicted $a_\mathfrak{p}$ table including the flagship $a_\mathfrak{p} = 0$ at smallest split prime $\mathfrak{p} \mid 19$, Ramanujan bound, inert-prime p^{-s} cancellation in $L(E_{11}/\mathbb{Q}(\sqrt{5}))$, and the scale separation from the ramified prime $p = 5$.
- `substrate-sim/algebra/onan_moonshine.py` — the six pariah sporadic groups with orders and prime factorisations, $|O'N|$ verification, pariah classes A and B, initial coefficients $A_e(D)$ of the O'Nan moonshine identity series with $A_e(-11) = 1$, Shimura-correspondence target prediction $\text{Sh}(Z_e) \propto f_{E_{11}}$ at level 11.

- `substrate-sim/algebra/tests/test_onan_moonshine.py` — 38 passing tests certifying: exactly 6 pariah groups, $|O'N| = 460\,815\,505\,920$ and its factorisation, pariahs containing $p = 11$ (J_1, J_4, Ly, O'N), Class A vs B partition (J_1 and O'N are Class A), small O'Nan moonshine coefficients including $A_e(-11) = 1$, Shimura target dimension 1 at level 11 (= genus of $X_0(11)$), and the pariah-count = spectral-immersion-depth agreement to 1%.
- `substrate-sim/algebra/tests/test_hopf_seam.py` — 41 passing tests certifying the Hopf-seam framework: nine Heegner numbers and CM j-invariants; $j(E_{11}) = -4096/11$ is not a CM j-invariant; minimal polynomials $\varphi^2 - \varphi - 1 = 0$ and $\omega_{-11}^2 - \omega_{-11} + 3 = 0$ at the same prime; Heegner ∩ Ogg = $\{2, 3, 7, 11, 19\}$; pariah Class A vs Class B partition; the five-fold convergence at $p = 11$.
- `substrate-sim/algebra/tests/test_minimality.py` — 27 passing tests certifying Propositions 7 and 8: the $(\mathbb{Q}(\sqrt{5}), \varphi, k = 5, p = 11)$ construction is the unique smallest-discriminant real-quadratic thin-group construction with non-trivial Lucas recurrence reaching an Ogg prime with positive-genus modular curve.
- `substrate-sim/algebra/hilbert_trace.py` (Phase 3 β) — Hilbert-modular dimension formula at level (11) giving $\dim S_{(2,2)}(\Gamma_0((11))) = 10$ (of which one is the base-change of $f_{E_{11}}$ and nine are genuinely-new Hilbert newforms), geometric/spectral partial-sum trace formula, and the factorisation residual at $s \in \{3, \ldots, 10\}$.
- `substrate-sim/algebra/tests/test_geodesic_hecke.py` (Phase 3 α) — 55 tests certifying: $\sqrt{5} \bmod p$ Tonelli–Shanks at split primes; Frobenius reduction at $\mathfrak{p}$ with the consistency relation $t \cdot \bar{t} \equiv N_{K/\mathbb{Q}}(a + b\varphi) \pmod{p}$; chi-squared discrimination between $a_{\mathfrak{p}} = 0$ primes ($\mathfrak{p} \mid 19$, $\mathfrak{p} \mid 29$, normalised $\chi^2 < 1$) and $|a_{\mathfrak{p}}| > 5$ primes ($\mathfrak{p} \mid 31$, $\mathfrak{p} \mid 41$, normalised $\chi^2 > 1$).
- `substrate-sim/algebra/pari_crossvalidate.py` (Phase 3 ε) — PARI/GP-based verification of every $a_p(E_{11})$ at primes ≤ 97. Found and fixed eight previously incorrect entries in the hardcoded table.
- `substrate-sim/algebra/pariah_depth.py` (Phase 3 γ) — Three independent structural attempts at $6 = \lceil 20 \log_5 \varphi \rceil$ (weight-1 cusp forms, Niemeier-Ogg lattice count, Hopf Chern class). All three fail to give a structural identification; the agreement is downgraded to "intriguing 0.4% coincidence."

Test suite totals after Phase 3. 741 passing tests across the algebra suite (up from 586 post-Phase-2): Phase 3 added 155 new tests across new modules (`test_hilbert_trace.py` 33, `test_geodesic_hecke.py` 55, `test_pari_crossvalidate.py` 35, `test_pariah_depth.py` 16, `test_phase3_catalog` plus 6 new tests in `test_onan_moonshine.py` for the Shimura proportionality theorem). The catalog explicitly maps every Phase 3 theorem T1–T5 and every open conjecture to at least one certifying test method.

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